

Yijiahao (Friedrich) Qi

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ABOUT ME

Senior undergraduate in Applied Physics, EECS Department, Peking University, advised by Prof. Meng Li. Currently an undergraduate research intern in the SHARC Lab, ECE Department, Georgia Tech, working with Prof. Cong (Callie) Hao on efficient Mixture-of-Agents (MoA) systems. Intending to apply for Ph.D. programs starting in Fall 2026.

Research Interest:

1. **Hardware-aware System Scheduling** to deploy modern machine learning algorithms onto edge devices, such as embodied agents, drones, and autonomous vehicles. *Related work:* CREATE (ASPLOS'26), DySL-VLA (DAC'26)
2. **Algorithm-tailored Hardware Reconfiguration** to maximize on-board resources. *Related work:* SpMM ASIC
3. **Efficient Multi-agent System** to rival server-level large models with efficient collaboration within powerful smaller models. *Related work:* Faster-MoA (DAC'26)

Hands-on Experience:

1. **Application (Efficient Embodied Agent):** a. Efficient Fault-tolerant Embodied-agent (CREATE) b. Robotic Manipulation Acceleration through Dynamic Layer-skipping (DySL-VLA).
2. **Algorithm:** a. PEFT small models via LLaMA-Factory b. Design customized metric for dynamic early exit.
3. **System:** a. KV Cache management and transmission b. Rrefilling and decoding flow.
4. **Hardware:** Sparse Matrix-matrix Multiplication (SpMM) Accelerator/Systolic Array (SA) Design based on Verilog.

EDUCATION

Peking University

Bachelor of Applied Physics; EECS Department; GPA: 3.75/4.0 (Top 18%);

Georgia Institute of Technology

Undergraduate Research Intern; Sharc Lab@ECE Department

Core Courses Covered:

- o Hardware Foundations of Artificial Intelligence (A+)
- o Machine Learning for Electronics Information Engineering (A)
- o Undergraduate Research Practice (A)
- o Chip Design using High-level Programming Language (A)

Beijing, China

Sep 2022 - Jul 2026

Atlanta, USA

Mar 2025 - Feb 2026

TECHNICAL SKILLS & TEST SCORES

Programming language: C++, Python, Pytorch, Verilog

Developer Tools: Vivado, Vitis, Linux, Git, Docker

TOEFL 113(120): Reading 30(30), Listening 28(30), Speaking 27(30), Writing 28(30)

GRE 329(340): Verbal Reasoning 159(170,80%), Quantitative Reasoning 170(170,91%), Analytical Writing 4.0(6.0,63%)

PUBLICATIONS

1. T. Xie*, **Y. Qi***, J. Wen, Z. Wan, Y. Dong, Z. Wang, S. Cai, Y. Liang, T. Jia, Y. Wang, R. Wang, and M. Li, "CERATE: Cross-layer Resilience Characterization and Optimization for Efficient yet Reliable Embodied AI Systems", accepted by the International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), 2026.
2. Z. Wang*, **Y. Qi***, H. Chen, Z. Wan, "Faster-MoA: Low-Latency Tree-Structured MoA Serving with Early Exit and Agent-Aware Prefill-Decode Overlap", accepted by the Design Automation Conference (DAC), 2026.
3. Z. Yang, **Y. Qi**, T. Xie, B. Yu, S. Liu, and M. Li, "DySL-VLA: Efficient Vision-language-action Model Inference via Dynamic-static Layer-skipping for Robot Manipulation", accepted by the Design Automation Conference (DAC), 2026.

RESEARCH PROJECTS

1. **Reliability assessment of LLM-based embodied AI system (*CREATE: Cross-Layer Resilience Characterization and Optimization for Efficient yet Reliable Embodied AI Systems*)**, accepted by *ASPLOS'26*, paper link: [CREATE](#)
Duration: Jul 2024 ~ Mar 2025 in PKU-SEC-Lab with Prof. Meng Li
Purpose:
 - o Access reliability of modern LLM-based embodied AI system.
 - o Propose error detection and correction methodologies at application, system, circuit level.**What I do:**
 - o Apply SmoothQuant's fault-injection methodology, identifying `o_proj` and `down_proj` as the most sensitive layers.
 - o Design a lightweight AD circuit that protects the planner, improving robustness from 10^{-7} to 10^{-3} BER.
 - o Use QuaRot to remap sensitive weight matrices to a more uniform domain, boosting robustness from 10^{-7} to 10^{-5} BER.
 - o Integrate an entropy-aware LDO for dynamic voltage scaling, cutting compute energy by 40.6%.**Experience gained:**
 - o **Algorithm:** a. LLM quantization b. Fault-tolerance profiling c. Error detection and correction skills.
 - o **Hardware:** Hardware architecture for embodied system.
2. **Mixture-of-Agents (MoA) Acceleration (*Faster-MoA: Low-Latency Tree-Structured MoA Serving with Early Exit and Agent-Aware Prefill-Decode Overlap*)**, accepted by *DAC'26*, paper link: [Faster-MoA](#)
Duration: Mar 2025 ~ Nov 2025 in GaTech-SHARC-Lab with Prof. Cong (Callie) Hao
Purpose: Rival server-level large models with small but powerful models through cross-agent communication.
What I do:
 - o Systematically profile the influence of different configuration on the overall performance of MoA system.
 - o Introduce a novel tree architecture for local aggregation.

- Prefilling and Decoding stage pipeline.
- Apply PEFT via LLaMA-Factory to diversify proposer exploration directions and decrease mutual semantic overlap.
- KV-Cache management for consecutive layers in MoA system.
- Use semantic similarity based early-exit to prune unnecessary inferences ($10 \times$ faster) with only $\pm 1\%$ accuracy variation.

Experience gained:

- **Algorithm:** a. PEFT models with PPO via LLaMA-Factory b. Design customized metrics for early-exit
- **System:** a. Construction of multi-agent system b. KV Cache management and transmission
- **Hardware:** Fine-granularity operator-level pipeline in multi-agent system with DP partition on several GPUs.

3. **VLA Model Inference Acceleration (*DySL-VLA: Efficient Vision-Language-Action Model Inference via Dynamic-Static Layer-Skipping for Robot Manipulation*)**, accepted by *DAC'26*, paper link: [DySL-VLA](#)

Duration: Feb 2025 ~ May 2025 in PKU-SEC-Lab with Prof. Meng Li

Purpose:

- Access inter-layer similarity of VLA model.
- Propose an adaptive layer-skip scheme to bypass different layers based on task requirements and motion significance.

What I do:

- Profile VLA models' robustness under dynamic layer skipping.
- Quantify activation similarity between adjacent layers (mean $\sim 80\%$)
- Train adapters and skip controllers in two stages, reducing trainable parameters by $85.7\times$ vs. full-parameter finetuning.
- Figure out the trajectory continuity can reflect the importance of the current action.

Experience gained:

- **Algorithm:** Finetuning of the VLA model with customized MLP modules.
- **System:** a. VLA model's layer-wise modification b. Robotic arm familiarization and action generation paradigm.

COURSE PROJECTS

1. **CircuitNet — Congestion Prediction for VLSI Layouts**, *Machine Learning for Electronics Information Engineering* final project, github link: [CircuitNet](#)

Purpose:

- Forecast cell-level routing congestion from post-placement layout images.
- Explore UNet-style architectures and low-bit quantization for accurate and efficient models in EDA tools.

What I do:

- Implement baseline UNet, deeper UNet, UNet+SE, Double-UNet and train on the CIRCUITNET dataset.
- Benchmark each model with NRMSE, SSIM and EMD.
- Perform ablation studies on depth, attention, normalization and activation choice.
- Run PTQ and QAT experiments of different bit-width on weights *and* activations.
- Analyze decoder- versus encoder-sensitivity to quantization noise.

Experience gained:

- Hands-on practice with VLSI-specific data preprocessing, tiling and augmentation.
- Deep dive into PTQ vs. QAT workflows, including custom bit-width and metric dashboards.
- Insights into the trade-off among model complexity, inference latency and predictive accuracy.

2. **Sparse Matrix-Matrix (SpMM) multiplication acceleration ASIC design**, *Chip Design using High-level Programming Language* final project, github link: [SpMM](#)

Purpose: Design an SpMM accelerator to optimize multiplication between an $N \times N$ row-compressed sparse matrix and an $N \times N$ dense matrix.

What I do:

- Implemented the FAN-network within the Reduction Unit (RedUnit) to efficiently compute partial sums of power-of-2 elements.
- Developed a PE core that converts received row-compressed data into a format compatible with RedUnit and performs Sparse Matrix-Vector (SpMV) multiplication, including cross-boundary sum calculations using a Halo-Adder.
- Designed a PE array architecture that transforms received ordinary matrices into row-compressed format and parallelizes SpMV computations into SpMM operations.
- Integrated double input/output buffer (Dbbuf), increasing the overall throughput by allowing simultaneous data transfer and computation.
- Realized output/weight stationary (os/ws) to support finer control while alleviating I/O occupation.

Experience gained:

- Hierarchical hardware design.
- Hardware optimization schemes like os/ws and Dbbuf.
- Absorbing state-of-the-art architecture from newly published papers.